Lecture 7

Variational approach to stochastic partial differential equations

3.3. The variational approach to stochastic partial differential equations

Let $(W_t)_t$ be an U-valued Q-Wiener process, where U is a separable real Hilbert space. Let V, H be as in the previous section. The following set of assumptions will set up the basic framework for our analysis.

Monotone-coercive SPDEs

Let $A : V \longrightarrow V'$ and $B : V \longrightarrow L(U, H)$ satisfy the following conditions:

(V.1) $\exists \alpha > 0, \lambda, \nu$ such that

$$2\langle A(u), u\rangle + \|B(u) \circ \sqrt{Q}\|^2_{L_2(U,H)} \leq -\alpha\|u\|^2_V + \lambda\|u\|^2_H + \nu \quad \forall u \in V.$$

(V.2) $\exists \lambda > 0$ such that

$$2\langle A(u) - A(v), u - v\rangle + \|(B(u) - B(v)) \circ \sqrt{Q}\|^2_{L_2(U,H)} \leq \lambda\|u - v\|^2_H \quad \forall u, v \in V.$$

(V.3) $\exists M > 0$ such that

$$\|A(u)\|_{V'} \leq M\,(1 + \|u\|_V) \quad \forall u \in V.$$

(V.4) $\forall u, v, w \in V$ such that the mapping

$$\lambda \mapsto \langle A(v + \lambda u), w\rangle$$

is continuous.

Given A and B as above, we want to solve the following stochastic evolution equation

$$du(t) = A(u(t))dt + B(u(t))dW(t), \quad u(0) = u_0. \tag{3.7}$$

The following theorem is the main result of the variational approach to SPDE:

THEOREM 3.5. *Under the assumptions (V.1) - (V.4) there exists for all $u_0 \in H$ a uniquely determined adapted process $u(t)$, $t \geq 0$, such that $u \in L^2([0, T]; V) \cap C([0, T]; H)$ and satisfying the equation*

$$\langle u(t), v\rangle = \langle u_0, v\rangle + \int_0^t \langle A(u(s)), v\rangle ds + \int_0^t \langle v, B(u(s))dW(s)\rangle.$$

For the proof we will need the following version of Itô's lemma:

LEMMA 3.6. *Let $u_0 \in H$, u (resp. v) be an adapted process with paths in $L^2([0, T]; V)$ (resp. $L^2([0, T]; V')$) and let M be a continuous H-valued local martingale, such that*

$$u(t) = u_0 + \int_0^t v(s)ds + M_t.$$

Then:

(*i*) $u \in C([0,T];H)$ *a.s.*,

(*ii*) $\|u(t)\|_H^2 = \|u_0\|_H^2 + 2\int_0^t \langle v(s), u(s)\rangle ds + 2\int_0^t \langle u(s), dM_s\rangle + \langle M\rangle_t$ *for all* $t \in [0,T]$ *a.s.*

Proof. For all $k \in \mathbb{N}$ we have that

$$u_k(t) := \langle u(t), e_k\rangle = \langle u_0, e_k\rangle + \int_0^t \langle v(s), e_k\rangle ds + \underbrace{\langle M_t, e_k\rangle}_{=:M_t^k}$$

is a continuous real-valued local semimartingale, such that Itô's-formula now implies that

$$du_k^2(t) = 2\langle v(t), e_k\rangle u_k(t)dt + 2u_k(t)dM_t^k + d\langle M^k\rangle_t.$$

Summing up w.r.t. k yields

$$\begin{aligned}\|u(t)\|_H^2 &= \sum_{k=1}^\infty u_k^2(t)\\ &= \underbrace{\sum_{k=1}^\infty \langle u_0, e_k\rangle^2}_{=\|u_0\|_H^2} + 2\int_0^t \underbrace{\sum_{k=1}^\infty \langle v(s), e_k\rangle u_k(s)}_{=\langle v(s),u(s)\rangle}\, ds\\ &\quad + 2\int_0^t \underbrace{\sum_{k=1}^\infty u_k(s)dM_s^k}_{=\langle u(s),dM_s\rangle} + \underbrace{\sum_{k=1}^\infty \langle M^k\rangle_t}_{=\langle M\rangle_t}\end{aligned}$$

This proves (ii). For the proof of (i) note that $u \in C([0,T];V')$ (since $u_0 \in H \subset V'$ is continuous, $\int_0^t v(s)\,ds \in V'$ is continuous and $M_t \in H \subset V'$ continuous) and therefore it remains to show that $t \mapsto u(t)$ is continuous in H w.r.t. the weak topology, since (ii) implies that $t \mapsto \|u(t)\|_H$ is continuous.
To see the weak continuity, consider first some $h \in V$. Then

$$\langle u(t), h\rangle = \langle u_0, h\rangle + \int_0^t \langle v(s), h\rangle ds + \langle M_t, h\rangle$$

is obviously continuous. For arbitrary $h \in H$ choose an approximating sequence $(h_m)_m \subset V$, converging towards h in H. Then

$$\sup_{t\in[0,T]} |\langle u(t), h - h_m\rangle| \le \underbrace{\sup_{t\in[0,T]} \|u(t)\|_H}_{<\infty} \|h - h_m\|_H \overset{m\nearrow\infty}{\longrightarrow} 0.$$

Indeed, $\sup_{t\in[0,T]} \|u(t)\|_H < \infty$ since $u \in L^\infty([0,T];H)$ due to part (ii) pf the lemma. It follows that $\langle u(t), h_m\rangle \to \langle u(t), h\rangle$ uniformly w.r.t. t, hence

$$t \mapsto \langle u(t), h\rangle$$

is continuous too. □

We also need the following version of Itô's formula:

LEMMA 3.7. *Let $\Phi : H \longrightarrow \mathbb{R}$ be two time continuously Frechet differentiable with $\Phi'(u) \in V$ for all $u \in V$ and such that $u \mapsto \Phi'(u)$, $V \longrightarrow V$ is continuous w.r.t. the weak topology and $\|\Phi'(u)\|_V \leq M\,(1 + \|u\|_V)$. Then the following Itô-formula holds:*

$$\Phi(u(t)) = \Phi(u_0) + \int_0^t \langle v(s), \Phi'(u(s))\rangle ds + \int_0^t \langle \Phi(u(s)), dM_s\rangle + \frac{1}{2}\int_0^t tr(\Phi''(u(s))Q(s))\, ds.$$

Here, $Q(s)$ is the integrated covariance of M, i.e., the nonnegative semidefinite and symmetric linear operator defined by

$$\int_0^t \langle Q(s)h, g\rangle_H ds = \langle \langle M, h\rangle_\cdot, \langle M, g\rangle_\cdot\rangle_t.$$

The proof of the Lemma is an exercise.

PROOF OF THEOREM 3.5.

Uniqueness Let $u, v \in L^2([0,T]; V) \cap C([0,T]; H)$ be two adapted solutions of (3.7). Define the stopping times

$$\tau_n := \inf\{t \in [0,T] : \|u(t)\|_H^2 > n \text{ or } \|v(t)\|_H^2 \text{ or } \int_0^t \|u(s)\|_V^2 \vee \|v(s)\|_V^2 ds > n\}.$$

Then $\tau_n \overset{n\nearrow\infty}{\longrightarrow} T$ a.s. and Lemma 3.6, applied to $u - v$, yields:

$$\begin{aligned}
\|u - v\|_H^2 &= 2\int_0^t \langle A(u(s)) - A(v(s)), u(s) - v(s)\rangle ds \\
&\quad + 2\int_0^t \langle u(s) - v(s), (B(u(s)) - B(v(s)))dW(s)\rangle \\
&\quad + \int_0^t \|(B(u(s)) - B(v(s))) \circ \sqrt{Q}\|_{L_2(U,H)}^2 ds \\
&\overset{(V.2)}{\leq} \underbrace{\int_0^t \langle u(s) - v(s), (B(u(s)) - B(v(s))dW(s))\rangle}_{\text{martingale up to } \tau_n \text{ for all } n\in\mathbb{N}} + \lambda \int_0^t \|u(s) - v(s)\|_H^2 ds.
\end{aligned}$$

Taking expectations yields

$$\mathbb{E}(\|u(t\wedge\tau_n) - v(t\wedge\tau_n)\|_H^2) \leq \lambda \int_0^t \mathbb{E}(\|u(s\wedge\tau_n) - v(s\wedge\tau_n)\|_H^2) ds,$$

so that Gronwall's lemma now implies that

$$\|u(t\wedge\tau_n) - v(t\wedge\tau_n)\|_H = 0 \quad \text{a.s.}$$

Taking the limit $n \to \infty$ we obtain that

$$u(t) - v(t) \quad \text{a.s.}$$

which proves uniqueness.

The proof of **existence** is carried out in the following two lemmata. □

Again, let $V_n = \text{span}\{e_1, ..., e_n\}$.

LEMMA 3.8. *For all n there exists an adapted process $u_n \in C([0,T]; V_n)$, such that*

(3.8)
$$\langle u_n(t), e_k\rangle = \langle u_n(t), e_k\rangle = \langle u_0, e_k\rangle + \int_0^t \langle A(u_n(s)), e_k\rangle ds + \int_0^t B_k(u_n(s))dW^n(s)$$

where

- $B_k(u)h := \langle B(u)h, e_k\rangle$,
- $W^n(t) := \sum_{l=1}^{n} \langle W(t), f_l\rangle f_l$, *with (f_l) being a CONS of U.*

(W.l.o.g. let (f_l) be eigenvectors of Q with corresponding eigenvalues $\lambda_l \geq 0$.)

LEMMA 3.9. *The following a priori estimate holds:*
$$\sup_n \mathbb{E}\left(\sup_{t\in[0,T]} \|u_n(t)\|_H^2 + \int_0^T \|u_n(s)\|_V^2 ds\right) < \infty.$$

We will postpone the proof of both lemmata and first complete the proof of the existence part of Theorem 3.5.

Proof of existence part of Theorem 3.5

Lemma 3.9 implies, that the sequence $(u_n)_n$ is bounded in $L^2(\Omega; C([0,T]; H)) \cap L^2(\Omega_T; V)$. This implies, using assumption (V.2):

(i) $(A(u_n)_n$ is bounded in $L^2(\Omega_T; V') = L^2(\Omega; L^2([0,T]; V'))$,
(ii) $(B(u_n)\circ\sqrt{Q})_n$ is bounded in $L^2(\Omega_T; L_2(U,H)) = L^2(\Omega; L^2([0,T]; L_2(U,H)))$.

Hence there exists a subsequence again denoted with $(u_n)_n$, such that

(i) $u_n \longrightarrow u$ weakly in $L^2(\Omega; L^2([0,T]; V))$,
(ii) $A(u_n) \longrightarrow g$ weakly in $L^2(\Omega; L^2([0,T]; V'))$,
(iii) $B(u_n) \circ \sqrt{Q} \longrightarrow \xi \circ \sqrt{Q}$ weakly in $L^2(\Omega; L^2([0,T]; L_2(U,H)))$.

We can take the limit $n \to \infty$ in (3.8) and deduce from this
$$\langle u(t), e_k\rangle = \langle u_0, e_k\rangle + \int_0^t \langle g(s), e_k\rangle ds + \int_0^t \langle e_k, \xi \circ \sqrt{Q}\rangle dW(s)$$

It remains to prove that

LEMMA 3.10. *$g = A(u)$ and $\xi \circ \sqrt{Q} = B(u) \circ \sqrt{Q}$.*

PROOF OF LEMMA 3.8. We can rewrite the integral equation (3.8) as a finite-dimensional stochastic differential equation

$$d\tilde{u}_n(t) = \widetilde{A}(\tilde{u}_n(t))dt + \widetilde{B}(\tilde{u}_n(t))d\widetilde{W}^n(t) \tag{3.9}$$

where

- $u_n(t) = i_n(\tilde{u}_n(t)$ and

$$i_n : \mathbb{R}^n \longrightarrow V_n \subset V,\ \tilde{u} \mapsto \sum_{k=1}^{n} \tilde{u}_k e_k$$

  denotes the natural embedding,
- $\widetilde{A}(\tilde{u}) = \pi_n A(i_n \tilde{u})$ and

$$\pi_n : V \longrightarrow \mathbb{R}^n,\ u \mapsto (\langle u, e_k \rangle)_{1 \le k \le n}$$

  denotes the natural orthogonal projection,
- $\widetilde{B}(\tilde{u}) = (B_k(i_n \tilde{u}))_{k=1,\dots,n}$,
- $\widetilde{W}^n(t) = (\langle W(t), f_l \rangle)_{l=1,\dots,n}$ for some CONS $(f_l)_l$ of U.

Clearly, $\widetilde{A}$ and $\widetilde{B}$ are continuous and

$$\begin{aligned}
&2\langle \widetilde{A}(\tilde{u}_1) - \widetilde{A}(\tilde{u}_2), \tilde{u}_1 - \tilde{u}_2 \rangle_{\mathbb{R}^n} + \|(\widetilde{B}(\tilde{u}_1) - \widetilde{B}(\tilde{u}_2)) \circ \sqrt{Q}\|^2_{\mathbb{R}^n} \\
&\quad = 2\langle A(i_n \tilde{u}_1) - A(i_n \tilde{u}_2), i_n \tilde{u}_1 - i_n \tilde{u}_2 \rangle + \|(B(i_n \tilde{u_1}) - B(i_n \tilde{u}_2)) \circ \sqrt{Q}\|^2_{L_2(U,H)} \\
&\quad \le \lambda \|i_n(\tilde{u}_1 - \tilde{u}_2)\|^2_H \\
&\quad = \lambda \|\tilde{u}_1 - \tilde{u}_2\|^2_{\mathbb{R}^n}
\end{aligned}$$

and similarly,

$$\begin{aligned}
2\langle \widetilde{A}(\tilde{u}), \tilde{u} \rangle_{\mathbb{R}^n} + \|\widetilde{B}(\tilde{u}) \circ \sqrt{Q}\|^2_{\mathbb{R}^n} &\le -\alpha \|i_n \tilde{u}\|^2_V + \lambda \|i_n \tilde{u}\|^2_H + \nu \\
&\le \lambda \|i_n \tilde{u}\|^2_H + \nu.
\end{aligned}$$

Now existence and uniqueness of the (probabilistically) strong solution to (3.9) (and therefore also to (3.8)) follows from standard results on solutios to stochastic differential equations (see, e.g. Wahrscheinlichkeitstheorie III). □

Proof of Lemma 3.9.

Applying Itô's formula to(3.8) yields

$$\begin{aligned}
\langle u_n(t), e_k \rangle^2_H = \langle u_0, e_k \rangle^2_H &+ 2 \int_0^t \langle A(u_n(s)), e_k \rangle \langle u_n(s), e_k \rangle ds \\
&+ 2 \int_0^t \langle u_n(s), e_k \rangle B_k(u_n(s)) dW^n(s) \\
&+ \sum_{l=1}^{n} \int_0^t B_k(u_n(s)) (\sqrt{Q} f_l)^2 ds.
\end{aligned}$$

Summing up w.r.t. $k = 1, \dots, n$ yields

$$\begin{aligned}\|u_n(t)\|_H^2 = \underbrace{\sum_{k=1}^n \Big(\langle u_0, e_k\rangle^2}_{\leq \|u_0\|_H^2} &+ 2\int_0^t \langle A(u_n(s)), e_k\rangle \langle u_n(s), e_k\rangle ds \\ &\underbrace{+ 2\int_0^t \langle u_n(s), e_k\rangle B_k(u_n(s)) dW^n(s)}_{2\int_0^t \langle u_n(s), B(u_n(s)) dW^n(s)\rangle} \\ &\underbrace{+ \sum_{l=1}^n \int_0^t B_k(u_n(s))(\sqrt{Q} f_l)^2 ds}_{\leq \|B(u_n(s))\circ\sqrt{Q}\|^2_{L_2(U,H)}} \Big).\end{aligned} \tag{3.10}$$

Indeed, recall that $B_k(u)h := \langle B(u)h, e_k\rangle$, $k = 1, 2, \dots, n$ and in the following write

$$B_{k,l}(u) := \langle B(u) f_l, e_k\rangle, k, l = 1, 2, \dots, n.$$

Then

$$B_k(u) dW^n(s) = \sum_{l=1}^n B_{k,l}(u) d \underbrace{\langle W(s), f_l\rangle}_{=:\beta_s^l}$$

and (β_s^l) are 1-dimensional Brownian motions with covariance

$$d\langle \beta_s^{l_1}, \beta_s^{l_2}\rangle = \langle Q f_{l_1}, f_{l_2}\rangle_U \, ds = \|\sqrt{Q} f_{l_1}\|^2 \delta_{l_1,l_2} \, ds,$$

since f_l are eigenvectors of Q. Consequently, the Itô corrections term for $\langle u_n(t), e_k\rangle^2$ has the form

$$\begin{aligned}&\sum_{l_1,l_2} \int_0^t B_{k,l_1}(u_n(s)) B_{k,l_2}(u_n(s)) \|\sqrt{Q} f_{l_1}\|^2 \delta_{l_1,l_2} \, ds \\ &\quad = \sum_{l=1}^n \int_0^t B_{k,l}(u_n(s))^2 \|\sqrt{Q} f_l\|^2 \, ds.\end{aligned}$$

Summing up w.r.t. $k = 1, 2, \dots, n$ yields the Itô correction term for $\|u_n(t)\|^2$:

$$\begin{aligned}&\sum_{k=1}^n \sum_{l=1}^n \int_0^t B_{k,l}(u_n(s))^2 \|\sqrt{Q} f_l\|^2 \, ds \\ &\quad = \sum_{l=1}^n \int_0^t \underbrace{\|B^n(u_n(s)) f_l\|_H^2 \|\sqrt{Q} f_l\|^2}_{=\|B^n(u_n(s))\circ\sqrt{Q} f_l\|_H^2} \, ds \\ &\quad \leq \int_0^t \|B^n(u_n(s)) \circ \sqrt{Q}\|^2_{L_2(U,H)} \, ds \\ &\quad \leq \int_0^t \|B(u_n(s)) \circ \sqrt{Q}\|^2_{L_2(U,H)} \, ds.\end{aligned}$$

Taking expectations and using (V.1) we obtain that

$$\begin{aligned}\mathbb{E}(\|u_n(t)\|_H^2) &\leq \|u_0\|_H^2 - \alpha \int_0^t \mathbb{E}(\|u_n(s)\|_V^2)ds + \lambda \int_0^t \mathbb{E}(\|u_n(s)\|_H^2)ds + \nu t \\ &\leq \|u_0\|_H^2 + \nu T + \lambda \int_0^t \mathbb{E}(\|u_n(s)\|_H^2)ds.\end{aligned}$$

Gronwall's lemma now implies that

$$\mathbb{E}(\|u_n(t)\|_H^2) \leq (\|u_0\|_H^2 + \nu T)e^{\lambda t} \leq (\|u_0\|_H^2 + \nu T)e^{\lambda T}.$$

It follows that

$$\begin{aligned}\alpha \int_0^T \mathbb{E}(\|u_n(s)\|_V^2)ds &\leq \|u_0\|_H^2 + \nu T + \lambda \int_0^T \mathbb{E}(\|u_n(s)\|_H^2)ds \\ &\leq \|u_0\|_H^2 + c_{T,\|u_0\|_H,\lambda,\nu}.\end{aligned}$$

The upper bound on the right hand side is independent of n which proves the boundedness of the second term.

In order to control the first term $\mathbb{E}\left(\sup_{t\in[0,T]} \|u_n(t)\|_H^2\right)$ we have to derive an upper bound of

$$\sup_{t\in[0,T]} e^{-\lambda t}\|u_n(t)\|_H^2 \leq \|u_0\|_H^2 + \sup_{t\in[0,T]} e^{-\lambda t}\nu t + \sup_{t\in[0,T]} \int_0^t e^{-\lambda s}\langle u_n(s), B(u_n(s))dW^n(s)\rangle.$$

But, the Burkholder-Davis-Gundy inequality implies

$$\begin{aligned}&\mathbb{E}\left(\sup_{t\in[0,T]} \int_0^t e^{-\lambda s}\langle u_n(s), B(u_n(s))dW^n(s)\rangle\right) \\ &\quad \leq \text{const. } \mathbb{E}\left(\left(\int_0^T \sum_{l=1}^n e^{-2\lambda s}\langle u_n(s), B(u_n(s)) \circ \sqrt{Q} f_l\rangle^2 ds\right)^{\frac{1}{2}}\right) \\ &\overset{\text{C.S.}}{\leq} \text{const. } \mathbb{E}\left(\sup_{t\in[0,T]} e^{-\frac{\lambda t}{2}}\|u_n(t)\|_H \left(\int_0^T e^{\lambda(T-s)}\|B(u_n(s)) \circ \sqrt{Q}\|_{L_2(U,H)}^2 ds\right)^{\frac{1}{2}}\right) \\ &\overset{\text{Young}}{\leq} \frac{1}{2}\mathbb{E}\left(\sup_{t\in[0,T]} e^{-\lambda t}\|u_n(t)\|_H^2\right) + \frac{\text{const.}^2}{2} \underbrace{\mathbb{E}\left(\int_0^T e^{\lambda(T-s)}\|B(u_n(s)) \circ \sqrt{Q}\|_{L_2(U,H)}^2 ds\right)}_{\overset{(V.3)}{\leq} \mathbb{E}(1+\int_0^T \|u_n(s)\|_V^2 ds}\end{aligned}$$

Hence

$$\mathbb{E}\left(\sup_{t\in[0,T]} \|u_n(t)\|_H^2\right) \leq \text{const.}$$

where the constant is independent of n! This proves the boundedness of the first term, hence the assertion. □

Proof of Lemma 3.10.

We apply again the "monotonicity-trick". To this end assume w.l.o.g. that (V.2) holds with $\lambda = 0$ (otherwise consider $A(u) - \lambda u$). Then

$$\int_0^T 2\langle A(u_n(s) - A(v(s)), u_n(s) - v(s)\rangle + \|(B(u_n(s)) - B(v(s)) \circ \sqrt{Q}\|^2_{L_2(U,H)} ds \leq 0$$

for all $v \in L^2([0,T];V)$. Weak convergence implies that

- $\int_0^T \langle A(u_n), v\rangle ds \longrightarrow \int_0^T \langle g, v\rangle ds$ for all $v \in L^2([0,T];V)$,
- $\int_0^T \langle A(v, u_n\rangle ds \longrightarrow \int_0^T \langle A(v), u\rangle ds$ for all $v \in L^2([0,T];V)$,
- $\int_0^T \langle B(u_n) \circ \sqrt{Q}, B(v) \circ \sqrt{Q}\rangle_{L_2(U,H)} ds \to \int_0^T \langle \xi \circ \sqrt{Q}, B(v) \circ \sqrt{Q}\rangle_{L_2(U,H)} ds$ for all $v \in L^2([0,T];V)$

weakly in $L^2(\Omega)$. Suppose we have in addition that

(3.11)
$$\mathbb{E}\left(\int_0^T 2\langle g(s), u(s)\rangle + \|\xi \circ \sqrt{Q}\|^2_{L_2(U,H)} ds\right) \leq \liminf_{n\to\infty} \mathbb{E}\left(\int_0^T 2\langle A(u_n(s)), e_n(s)\rangle ds + \int_0^T \|B(u_n(s)) \circ \sqrt{Q}\|^2_{L_2(U,H)} ds\right).$$

It then follows for all $v \in L^2(\Omega; L^2([0,T];V))$ that

$$\mathbb{E}\left(\int_0^T 2\langle g(s) - A(v(s)), u(s) - v(s)\rangle + \|(\xi - B(v(s)) \circ \sqrt{Q}\|^2_{L_2(U,H)} ds\right) \leq 0,$$

since

$$\begin{aligned}\mathbb{E}&\left(\int_0^T 2\langle g(s) - A(v(s)), u(s) - v(s)\rangle + \|(\xi - B(v(s)) \circ \sqrt{Q}\|^2_{L_2(U,H)} ds\right)\\ &\leq \liminf_{n\to\infty} \mathbb{E}\left(\int_0^T 2\langle A(u_n) - A(v), u_n - v\rangle ds + \int_0^T \|(B(u_n) - B(u)) \circ \sqrt{Q}\|^2_{L_2(U,H)} ds\right)\\ &\overset{(V.2)}{\leq} 0.\end{aligned}$$

Choosing first $v = u$ we obtain that

$$\mathbb{E}\left(\int_0^T \|\xi \circ \sqrt{Q} - B(u) \circ \sqrt{Q}\|^2_{L_2(U,H)} ds\right) \leq 0.$$

Since $\|\cdot\|_{L_2(U,H)} \geq 0$ this implies

$$\xi \circ \sqrt{Q} = B(u) \circ \sqrt{Q} \ \mathbb{P} - a.s.$$

But then

$$\mathbb{E}\left(\int_0^T \langle g(s) - A(v(s)), (u-v)(s)\rangle\, ds\right) \leq 0$$

for all $v \in L^2(\Omega; L^2([0,T];V))$. Choosing this time $v = u - \theta w$, with $\theta > 0$ and $w \in L^2(\Omega; L^2([0,T];V))$ arbitrary, yields

$$\theta\mathbb{E}\left(\int_0^T \langle g(s) - A(u(s) - \theta w(s)), w(s)\rangle ds\right) \leq 0.$$

Since A is hemi-continuous (due to (V.4)) we have that $\theta \mapsto \langle A(u - \theta w), w\rangle$ is continuous. Hence taking the limit $\theta \searrow 0$ implies that

$$\mathbb{E}\left(\int_0^T \langle g(s) - A(u(s)), w(s)\rangle ds\right) \leq 0.$$

Since w was arbitrary, we conclude that $g(s) = A(u(s))$ a.s.

It remains to establish inequality (3.10). To this end note that (3.9) implies that

$$\begin{aligned}\mathbb{E}&\left(\int_0^T 2\langle A(u_n(s)), u_n(s)\rangle + \|B(u_n(s)) \circ \sqrt{Q}\|^2_{L_2(U,H)} ds\right)\\ &\geq \mathbb{E}\left(\|u_n(T)\|^2_H\right) - \|u_0\|^2_H.\end{aligned}$$

Hence (3.10) follows from the inequality

$$\liminf_{n\to\infty} \mathbb{E}\left(\|u_n(T)\|^2_H\right) \geq \mathbb{E}\left(\|u(T)\|^2_H\right),$$

which is a consequence of the weak convergence of $u_n(T) \longrightarrow u(T)$ in $L^2(\Omega; H)$ along some subsequence. □

Example 3.11 (stochastic heat equation with multiplicative noise). Consider the stochastic bilinear heat equation

$$\partial_t u(t,x) = \frac{1}{2}\partial_{xx} u(t,x) + \theta \partial_x u(t,x)\partial_t W_t, \qquad u(0,x) = u_0(x)$$

with $\theta \in \mathbb{R}$. In this case

(1) $A(u) = \frac{1}{2}u_{xx} : H^1(\mathbb{R}) \subset L^2(\mathbb{R}) \simeq L^2(\mathbb{R})' \subset (H^1(\mathbb{R}))'$ and

$$_{(H^1)'}\langle A(u), u\rangle_{H^1} = \frac{1}{2}\int u_{xx} u dx = -\frac{1}{2}\int u_x u_x dx = -\frac{1}{2}\int u_x^2 dx = -\frac{1}{2}\|u\|^2_{H^1} + \frac{1}{2}\|u\|^2_H$$

(2) $B(u)h = \partial_x u h$ with

$$\|B(u) \circ \sqrt{Q}\|^2_{L_2(U,H)} = \|\partial_x u \circ \sqrt{Q}\|^2_{L_2(U,H)} \overset{U \equiv \mathbb{R}}{=} \|\partial_x u\|^2_{L^2(\mathbb{R})},$$

(3) (W_t) is a 1-dimensional Wiener process (hence standard Brownian motion), i.e. $U = \mathbb{R}$.

In this case the coercivity assumption (V.1)

$$2\langle A(u), u\rangle + \|B(u) \circ \sqrt{Q}\|^2_{L_2(U,H)} = (\theta^2 - 1)\|\partial_x u\|^2_{L^2(R)}$$

is satisfied if and only if $|\theta| < 1$.

In the case $\theta = 1$ we have that

$$u(t,x) := u_0(x + W_t)$$

is an explicit solution, in the case $\theta = -1$ we can consider $u(t,x) = u_0(x - W_t)$. Indeed, an applications of Ito's formula yields for $u_0 \in C^2(\mathbb{R})$

$$\begin{aligned} u_0(x + W_t) &= u_0(x) + \int_0^t \partial_x u_0(x + W_s) dW_s + \frac{1}{2} \int_0^t \partial_{xx} u_0(x + W_s) ds \\ &= u_0(x) + \int_0^t \partial_x u(s,x) dW_s + \frac{1}{2} \int_0^t u_{xx}(s,x) ds \end{aligned}$$

It is clear that $u(t,x) = u_0(x + W_t)$ has the same spatial regularity as u_0, hence there is no regularizing effect in the case and the equation becomes hyperbolic.

In the case $\theta = 0$ the spde reduces to the deterministic heat equation with explicit solution

$$u(t,x) = \frac{1}{\sqrt{2\pi t}} \int e^{-\frac{(x-y)^2}{2t}} u_0(y) dy$$

In the case $|\theta| > 1$ we obtain that

$$2\langle A(u), u\rangle + \|B(u) \circ \sqrt{Q}\|^2 = -(1 - \theta^2)\|\partial_x u\|_{L^2}^2$$

which corresponds to a backward heat equation which is no longer well-posed.

Can one detect these effects in numerical simulations?